

1. Data Preparation



Raw data

Pre-segmented Speech Clips from corpora



Labeld



Structured lable data

Lael Accent, emotion, style, etc.



Perturbed



Attribute-Perturbed data

Controlled pitch / speed / volume modifications

2. Controllable Diversification

Prompt

Rewrite the following TTS attributes into natural language instructions. You must adhere to specific constraints...



specific constraints



Syntactic



Persona



Attr. Slots



Generate instructions

Instructions

hey, can you use the voice of father...

3. Drift Filtering

Instructions

hey, can you use the voice of father...



LLM Verifier
Scoring & Voting



Drift Taxonomy

Instruction-to-Execution Drift

I'm furious! How dare...

Role-Assumption Drift

As a child, I would say...

Entity/Label Drift

[Emotion: Sad]->Use happy...



High-Fidelity
Instructions

4. Stabilized SFT & Evaluation

High-Fidelity
Instructions

Speech Clips



Instruct-TTS Model



InstructTTEval (CH)



TTS Synthesis



Eval



Human
True
False



Gemini
True
False

